

Brian Lam, Joshua Johns, Riley Matthiessen
Crab Team
Washington Sea Grant

4/14/2026

To:

Program Officers: José Guzmán, PhD, Megan Ewing, Sarah Tanja, Andy Nutzhorn
Funding Organization: International Society of Crustacean Ecophysiology
Program Title: *Applied Comparative Ecophysiology of Hemigrapsus oregonensis and Carcinus maenas: Linking Mechanisms to Management in a Changing Coastal Environment*

Subject: Letter of Intent to Submit a Proposal – *Applied Comparative Ecophysiology of Hemigrapsus oregonensis and Carcinus maenas*

Dear José Guzmán, PhD, Megan Ewing, Sarah Tanja, and Andy Nutzhorn,

I am writing to formally express my intent to submit a proposal in response to the RFP titled “*Applied Comparative Ecophysiology of Hemigrapsus oregonensis and Carcinus maenas: Linking Mechanisms to Management in a Changing Coastal Environment.*”

Our proposed project, tentatively titled "**The effects of *Carcinus maenas* presence on behavior and physiology of *Hemigrapsus oregonensis* over various temperatures**", will address the following thematic area(s) of the RFP:

- Presence of the invasive *Carcinus maenas*
- Applied Stress Physiology of *Hemigrapsus oregonensis*
- Heat Stress of *H. oregonensis*
- Feeding behavioral ecology

This project will investigate the behavioral and physiological effects of the predatory European green crab (*Carcinus maenas*) on hairy shore crabs (*Hemigrapsus oregonensis*) in various temperatures. This project is relevant to areas of conservation and fishery management because the impacts of invasive *C. maenas* can have large effects on shore crab populations and native crab fisheries.

The anticipated Principal Investigators will be Brian Lam, Joshua Johns, and Riley Matthiessen. Our research approach emphasizes ecologically meaningful physiological endpoints with direct application to coastal resource management and climate adaptation.

We appreciate this opportunity to contribute to applied coastal science and would be happy to provide any additional information if needed prior to full proposal submission.

Sincerely,
Brian Lam, Joshua Johns, and Riley Matthiessen